**Minor Project Synopsis Report**

# Predictive Maintenance for Autonomous Electric Buses using Real-time Sensor Data and Machine Learning

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# Index


| Sr.No. | Content | Page No. |
|---|---|---|
| 1 | Introduction | 1 |
| 2 | Motivation | 2 |
| 3 | Literature Review | 3-5 |
| 4 | Gap Analysis | 6 |
| 5 | Problem statement | 7 |
| 6 | Objectives | 8 |
| 7 | Tools and Technologies | 9 |
| 8 | Methodology | 10-11 |
| 9 | Conclusion | 12 |
| 10 | References | 13-14 |

# Abstract

The adoption of Autonomous Electric Buses (AEBs) is rapidly increasing due to thei significant environmental benefits, enhanced passenger safety, and their ability to improve the overall efficiency of modern urban transportation systems. Despite these advantages, the maintenance of such systems remains a major challenge because of the complex integration of mechanical components, electronic systems, and embedded control units. Traditional maintenance strategies, such as reactive and time-based approaches, are often inefficient and can lead to unexpected failures, increased downtime, and higher operational costs. To address these issues, this study focuses on the implementation of predictive maintenance techniques using real-time sensor data combined with advanced machine learning models, particularly Convolutional Neural Networks (CNNs), for accurate fault detection and system monitoring. The proposed system is built around three key components: the MPU6050 Inertial Measurement Unit (IMU), the piezoelectric vibration sensor, and the ESP32 microcontroller. The MPU6050 sensor plays a crucial role in continuously monitoring motion parameters such as acceleration and angular velocity, which helps in identifying abnormal movement patterns or instability in the vehicle. At the same time, the piezo sensor is used to detect vibrations generated by different mechanical parts, allowing the system to capture early signs of wear, imbalance, or potential mechanical failures. The ESP32 microcontroller serves as the central processing unit, efficiently collecting data from both sensors, performing initial preprocessing, and enabling real-time data transmission for further analysis. By leveraging CNN-based models, the system can process this high-dimensional sensor data to automatically extract meaningful features and detect anomalies with greater accuracy compared to traditional methods. This intelligent approach not only enables early fault prediction but also helps in minimizing maintenance costs, reducing unexpected breakdowns, and improving the overall reliability and lifespan of the system. Ultimately, the integration of these cost-effective components with advanced predictive algorithms contributes to the development of a robust, scalable, and efficient maintenance framework, supporting the advancement of smart and sustainable transportation systems.



**KEYWORDS:**

Predictive Maintenance, Autonomous Electric Buses, Convolutional Neural Networks (CNN), MPU6050, Piezo Sensor, ESP32, Fault Detection, Smart Transportation

# 1. Introduction

Rapid urbanization has placed increasing pressure on public transportation systems, creating a strong demand for solutions that are efficient, safe, and environmentally sustainable. Autonomous Electric Buses (AEBs) have emerged as a promising approach by combining electric mobility with intelligent automation, helping reduce emissions while improving operational efficiency and safety. However, the maintenance of such systems remains a significant challenge due to the complexity of their interconnected mechanical and electronic components. Traditional maintenance methods, such as reactive and time-based approaches, are often ineffective because they either address problems too late or rely on fixed schedules that do not reflect the actual condition of the system. To overcome these limitations, predictive maintenance (PdM) has gained attention as a data-driven approach that uses real-time monitoring and machine learning techniques to predict potential failures before they occur. In this context, the proposed system focuses on the use of three key components: the MPU6050 sensor, a piezoelectric vibration sensor, and the ESP32 microcontroller. The MPU6050 is used to continuously monitor motion parameters such as acceleration and angular velocity, enabling the detection of abnormal movement patterns, while the piezo sensor captures vibration signals that help identify early signs of mechanical faults or wear. The ESP32 acts as the central processing unit, efficiently collecting, processing, and transmitting sensor data in real time. By applying Convolutional Neural Networks (CNNs) to the data obtained from these components, the system can analyze complex patterns and detect anomalies with high accuracy. This approach provides a more reliable and efficient maintenance strategy, reducing downtime and operational costs while improving the safety and performance of autonomous electric bus systems, ultimately supporting the development of smart and sustainable urban transportation.

## 2. Motivation

As cities continue to expand and public transportation systems increasingly shift toward electric and autonomous technologies, the expectations for safety, reliability, and efficiency have grown significantly. Autonomous Electric Buses (AEBs) play a crucial role in modern urban mobility, serving a large number of daily commuters. However, any unexpected failure in these systems can lead to serious consequences such as service disruptions, increased traffic congestion, economic losses, and safety risks. The complexity of AEBs, arising from the integration of mechanical and electronic subsystems, makes maintenance a challenging task. Traditional maintenance approaches, such as reactive and preventive methods, are no longer sufficient because they either respond only after failures occur or rely on fixed schedules that do not reflect the actual condition of system components.

To overcome these limitations, predictive maintenance has emerged as a more advanced and intelligent solution that utilizes real-time sensor data and machine learning techniques to monitor system health and detect faults at an early stage. In this study, a practical and cost-effective approach is considered using three key components: the MPU6050 sensor, a piezoelectric vibration sensor, and the ESP32 microcontroller. The MPU6050 continuously monitors motion parameters such as acceleration and angular velocity, helping to identify abnormal movement patterns, while the piezo sensor detects vibrations that may indicate mechanical wear or faults. The ESP32 serves as the central unit, efficiently collecting, processing, and transmitting the sensor data in real time, enabling continuous system monitoring.

By applying Convolutional Neural Networks (CNNs) to the data collected from these sensors, the system can automatically extract meaningful features and detect hidden fault patterns with high accuracy. This data-driven approach enables early fault prediction, reduces unexpected breakdowns, minimizes maintenance costs, and improves overall system reliability. Moreover, it enhances passenger safety and ensures smoother operation of autonomous electric buses. Therefore, this study is motivated by the need to develop a simple yet effective predictive maintenance framework using accessible components, contributing to smarter, safer, and more sustainable urban transportation systems.

## 3. Literature Review

In recent years, Predictive Maintenance (PdM) has gained significant importance in electric vehicles and intelligent transportation systems as a more advanced alternative to traditional maintenance strategies. Unlike reactive maintenance, which only addresses failures after they occur, or preventive maintenance, which relies on fixed service intervals, predictive maintenance uses real-time sensor data and machine learning techniques to anticipate faults before they become critical. This shift toward condition-based monitoring has greatly improved system reliability and operational efficiency. In the context of this study, a practical and cost-effective approach is considered using three key components: the MPU6050 sensor, a piezoelectric vibration sensor, and the ESP32 microcontroller. The MPU6050 enables continuous monitoring of motion parameters such as acceleration and angular velocity, while the piezo sensor captures vibration patterns that indicate mechanical wear or abnormalities. The ESP32 acts as the central processing unit, handling real-time data acquisition, preprocessing, and communication, making the system efficient and scalable for predictive maintenance applications.

Machine learning techniques play a vital role in enabling predictive maintenance by analyzing sensor data and identifying hidden fault patterns. Traditional models such as Support Vector Machines (SVM) and Random Forests have been widely used for classification-based fault detection, while deep learning methods like Long Short-Term Memory (LSTM) networks are effective for time-series analysis and degradation prediction. However, these methods often require manual feature extraction or struggle with high-dimensional data. Convolutional Neural Networks (CNNs) have emerged as a powerful alternative because they can automatically learn meaningful features directly from raw sensor data. In this system, CNNs can process data collected from the MPU6050 and piezo sensors to detect anomalies in motion and vibration patterns with high accuracy. Their ability to capture complex relationships within multi-dimensional data makes them particularly suitable for predictive maintenance in systems where multiple parameters must be analyzed simultaneously.

Several studies in the literature support the effectiveness of predictive maintenance and machine learning techniques in improving system performance and reducing maintenance costs. Research has shown that data-driven approaches significantly enhance early fault detection and overall reliability compared to traditional methods. However, many existing works focus on individual components or rely on offline datasets, limiting their practical applicability. There is still a lack of simple, real-time, and system-level implementations that integrate multiple sensors into a unified framework. This study addresses these gaps by combining MPU6050-based motion analysis, piezo sensor-based vibration monitoring, and ESP32-based real-time processing with CNN-based fault detection. Such an approach not only simplifies implementation but also provides an efficient and scalable solution for improving reliability, safety, and operational efficiency in autonomous electric bus systems.

**Table 1: Literature Review of Predictive Maintenance Approaches**

| Author(s) | Technique Used | Key Focus Area | Key Contribution | Research Gap |
|---|---|---|---|---|
| **Jardine et al. (2006)** | Statistical PdM | Condition monitoring | Established foundation of predictive maintenance | No real-time sensor integration |
| **Widodo & Yang (2007)** | SVM | Fault diagnosis | Showed ML effectiveness in vibration analysis | Limited multi-sensor usage |
| **Lei et al. (2018)** | Data-driven ML | RUL prediction | Improved fault prediction accuracy | Lacks lightweight implementation |
| **Zonta et al. (2020)** | ML + IoT | Smart maintenance | Integrated PdM with IoT systems | Limited real-time edge processing |
| **Krupitzer et al. (2020)** | Survey | PdM architecture | Classified predictive maintenance frameworks | No deep learning comparison |
| **Zhao / Zhang et al. (2022)** | CNN | Fault detection | Demonstrated CNN effectiveness in pattern recognition | Limited multi-sensor integration |
| **Achouch et al. (2022)** | ML + Industry 4.0 | PdM challenges | Identified future research directions | High system complexity |

| | | | | |
|---|---|---|---|---|
| **Abidi et al. (2022)** | SVM + RNN | Fault prediction | Combined ML & DL models | Limited scalability |
| **CNN-based Study (2023)** | CNN | Fault classification | Improved accuracy using feature extraction | Needs real-time validation |
| **Deep Learning PdM (2024)** | CNN + Hybrid DL | Multi-sensor analysis | Enhanced performance using hybrid models | High computational complexity |
| **AI Smart Maintenance (2025)** | CNN + AI | Smart transportation | Focus on intelligent fleet systems | Limited real-world AEB studies |

## 4. Gap Analysis

Although significant research has been carried out in predictive maintenance and machine learning–based prognostics, several limitations still exist when applying these approaches to Autonomous Electric Bus (AEB) systems. Most existing studies focus on isolated components or controlled industrial environments, which do not fully represent the complexity of real-world transportation systems. In practical scenarios, multiple subsystems operate simultaneously, and their interactions make fault detection more challenging. Additionally, many studies rely on offline datasets rather than real-time sensor data, limiting their applicability in dynamic environments. There is also limited exploration of lightweight and cost-effective hardware implementations that can support real-time predictive maintenance. These gaps highlight the need for integrated, real-time, and system-level solutions using efficient components such as MPU6050 for motion sensing, piezo sensors for vibration analysis, and ESP32 for data processing. The use of Convolutional Neural Networks (CNNs) further provides an opportunity to improve fault detection accuracy by automatically extracting features from multi-sensor data.

**Table 2: Summary of major research gaps identified from the literature.**

| Area | Existing Research Focus | Identified Gap | Research Direction (CNN-Based) |
|---|---|---|---|
| **Application Domain** | Primarily industrial machinery and manufacturing systems | Limited research on autonomous electric bus systems | Need for domain-specific studies for AEBs |
| **System Monitoring** | Single-parameter or isolated monitoring | Lack of integrated monitoring using motion and vibration data | Multi-sensor data fusion using CNN |
| **Data Type** | Use of historical/offline datasets | Limited real-time data processing | CNN-based real-time data analysis using ESP32 |
| **Fault Detection** | Traditional ML (SVM, RF), threshold methods | Difficulty in detecting complex fault patterns | CNN-based automatic feature extraction |
| **Hardware Implementation** | High-cost and complex systems | Limited use of low-cost embedded systems | Use of MPU6050, Piezo sensor, and ESP32 |
| **Real-Time Processing** | Cloud-dependent architectures | Latency and dependency issues | Edge-based processing using ESP32 + CNN |

## 5. Problem Statement

The adoption of autonomous electric buses is rapidly increasing as cities aim to develop smarter and more sustainable transportation systems. These buses provide important advantages such as reduced emissions, improved operational efficiency, and enhanced safety. However, they also depend on complex and interconnected components, which makes maintenance a challenging task. Even a minor fault in the system can affect overall performance, leading to service disruptions, increased costs, and potential safety risks. Traditional maintenance approaches are not effective in such scenarios, as reactive maintenance only addresses issues after failure, while preventive maintenance follows fixed schedules that do not reflect the actual condition of the system. To overcome these limitations, this study focuses on a predictive maintenance approach using simple and cost-effective components such as the MPU6050 sensor, piezoelectric vibration sensor, and ESP32 microcontroller. The MPU6050 monitors motion and detects abnormal behavior, the piezo sensor captures vibration patterns indicating mechanical issues, and the ESP32 enables real-time data collection and processing.

Although predictive maintenance has been widely explored in industrial systems and electric vehicles, its application to autonomous electric buses remains limited. Most existing studies focus on individual components and rely on offline datasets, which reduces their effectiveness in real-world conditions. Key challenges such as real-time monitoring, multi-sensor data integration, and efficient fault detection are still not fully addressed. In this context, Convolutional Neural Networks (CNNs) offer a powerful solution due to their ability to automatically extract meaningful features from complex sensor data. By applying CNNs to the data collected from MPU6050 and piezo sensors, the system can detect faults more accurately and at an early stage. This approach helps improve reliability, reduce maintenance costs, and enhance overall system performance, making it highly suitable for modern autonomous electric bus systems.

## 6. Objectives

The primary objective of this study is to examine predictive maintenance techniques for Autonomous Electric Buses (AEBs) using real-time sensor data, with a focus on Convolutional Neural Networks (CNNs) for fault detection and system monitoring.

**Specific Objectives:**

- To analyze motion data from the MPU6050 IMU sensor for detecting abnormal acceleration and orientation patterns.
- To evaluate vibration data from the piezo sensor for early identification of mechanical faults.
- To utilize the ESP32 microcontroller for real-time data acquisition, processing, and communication.
- To integrate multi-sensor data using CNN-based techniques for accurate fault detection and improved system reliability.

# 7. Tools and Technologies

This study examines the role of modern software tools, programming environments, and data-driven technologies in enabling predictive maintenance for autonomous electric buses. The selected technologies support real-time data collection, storage, analysis, and machine learning–based fault detection using sensor data from MPU6050, piezo sensors, and ESP32.

- **Cloud platforms** such as Amazon Web Services and Microsoft Azure provide scalable infrastructure for handling large volumes of real-time sensor data. These platforms support data storage, remote monitoring, and centralized analysis of multiple vehicles, enabling efficient fleet management and system scalability.
- **MySQL** is used for structured storage of sensor data, including motion and vibration readings, along with maintenance records. It ensures data integrity, efficient querying, and organized storage for both real-time and historical analysis.
- **Python** is used as the primary programming language due to its flexibility and strong support for data processing and machine learning. It allows easy integration with sensors, databases, and analytical tools.
- **NumPy** and **Pandas** are used for efficient data handling and preprocessing. NumPy enables fast numerical computations, while Pandas supports data cleaning, transformation, and structuring of sensor data for analysis.
- Frameworks such as **TensorFlow, Keras,** and **Scikit-learn** are used to develop predictive models. These tools support tasks like fault detection, classification, and performance evaluation using sensor data.
- **Convolutional Neural Networks (CNNs)** are applied for analyzing motion and vibration data collected from MPU6050 and piezo sensors. CNNs automatically extract features from sensor data, enabling accurate detection of complex fault patterns. Their integration with ESP32-based real-time data acquisition improves system reliability and predictive accuracy.

# 8. Methodology

The methodology adopted in this study follows a systematic data-driven approach for implementing predictive maintenance in autonomous electric buses using multi-sensor inputs, including IMU (MPU6050), vibration (piezo sensor), and camera-based data. The process is divided into multiple phases, covering data acquisition, preprocessing, feature extraction, model prediction, and decision fusion to accurately detect faults and improve system reliability.

**Phase 1: Data Acquisition**

The first phase involves collecting real-time data from multiple sensors integrated into the system. The MPU6050 IMU sensor captures motion parameters such as acceleration and orientation, while the piezo sensor records vibration signals that reflect mechanical behavior. In addition, a camera is used to capture visual data for identifying external or structural anomalies. These sensors generate continuous data streams representing system performance under both normal and faulty conditions, enabling early detection of abnormal patterns.

**Phase 2: Data Preprocessing**

The collected raw data may contain noise, missing values, and inconsistencies. Therefore, preprocessing techniques such as filtering, normalization, and data cleaning are applied to improve data quality. IMU and vibration signals are smoothed and normalized, while camera data is resized and formatted for efficient processing. This phase ensures that all sensor inputs are consistent and suitable for further analysis.

**Phase 3: Feature Extraction and Learning**

Feature extraction is performed differently for sensor data and image data. For IMU and vibration sensors, statistical and signal-based features such as mean, variance, and frequency patterns are extracted to represent system behavior. For camera data, Convolutional Neural Networks (CNNs) are used to automatically learn and extract important visual features. CNNs eliminate the need for manual feature engineering and are highly effective in identifying complex patterns from image inputs.

**Phase 4: Model Prediction**

In this phase, the processed data is fed into predictive models for fault detection. Sensor-based features and CNN-extracted image features are analyzed to classify the system condition as normal or faulty. CNN models play a crucial role in identifying hidden patterns and anomalies with high accuracy, especially in multi-dimensional data.

**Phase 5: Decision Fusion**

The outputs obtained from different sources (IMU, vibration, and camera) are combined in this phase. Decision fusion techniques integrate results from sensor-based and CNN-based models to produce a unified and more reliable prediction. This helps in reducing false alarms and improving overall detection accuracy.

**Phase 6: Predictive Monitoring and Final Output**

The final phase provides the system output in terms of "Normal" or "Fault Detected." Continuous monitoring ensures early fault detection and supports maintenance decision-making. This integrated approach enhances system reliability, reduces downtime, and improves safety in autonomous electric bus operations.

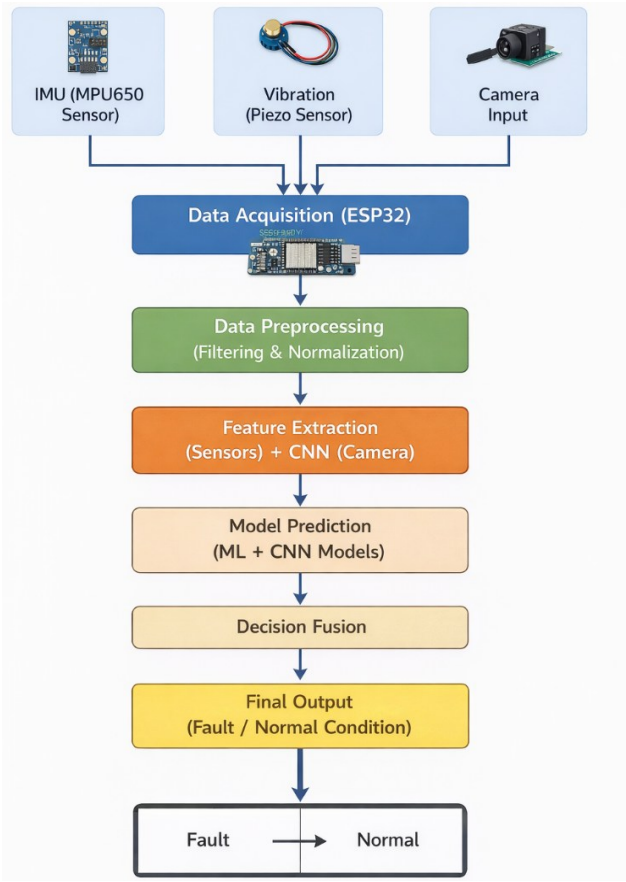


Figure 1: Predictive Maintance System

# 9. Conclusion

The increasing adoption of autonomous electric buses in urban transportation systems highlights the need for more advanced and reliable maintenance strategies, as traditional reactive and preventive approaches are no longer sufficient to manage the complexity of modern electric and autonomous systems. This study emphasizes predictive maintenance as an effective, data-driven solution that leverages real-time sensor data from components such as the MPU6050 IMU sensor, piezoelectric vibration sensor, and ESP32 microcontroller to monitor system behavior and detect early signs of faults. By analyzing motion and vibration patterns, the system enables timely identification of abnormalities, reducing the risk of unexpected failures. Machine learning and deep learning techniques, particularly Convolutional Neural Networks (CNNs), further enhance this approach by accurately detecting anomalies and improving fault prediction capabilities. The integration of multi-sensor data with real-time processing not only increases system reliability and efficiency but also reduces maintenance costs and downtime. Additionally, this approach improves passenger safety and supports better maintenance decision-making. Overall, the study demonstrates the importance of combining cost-effective hardware with intelligent algorithms to develop scalable and efficient predictive maintenance systems, contributing to smarter, safer, and more sustainable urban transportation, while future advancements in deep learning are expected to further enhance performance and real-time applicability.

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